

# Proto-Spinor Closure and Worldsheet Encoding in Modal Triplet Theory

Corrected fourth edition: a conditional local bridge

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## Abstract

We formulate the proto-spinor/worldsheet relation as a typed local bridge rather than a derivation of string theory. Near an aligned q79-compatible background, a bridge map sends proto-spinor and closure-strain perturbations to worldsheet coupling perturbations. If its derivative intertwines the two quadratic forms and its nonlinear remainder is controlled, the two descriptions have the same quadratic admissibility test on the selected subspace. We give the precise domain, codomain, error estimate, and block conditions. Circle, lens, and nil label shared phase, finite transport, and anchoring blocks; they are not assumed to be literal product factors. Weyl, Dirac, Majorana, and twistor languages remain conditional charts. Worldsheet Weyl invariance, modular consistency, anomaly cancellation, ghosts, and target-space equations are independent gates. The selected q79 trace carrier supplies a concrete global target for the bridge, but the same-source connection intertwiner is still required.

## Revision note for this edition

**Supersedes.** *Proto-Spinor Closure and Worldsheet Encoding in Modal Triplet Theory*, version 3.

**Reason.** The shadow correspondence lacked one explicit source-to-target map, a declared validity ball, a nonlinear error estimate, and separation from full worldsheet consistency.

**Resolution.** Version 4 defines a  $C^2$  bridge map, proves Hessian intertwining implies quadratic diagnostic agreement with cubic error, and lists the independent Weyl, modular, ghost, anomaly, and beta-function gates.

**Retained result.** The earlier blockwise matching survives as a conditional local bridge near alignment.

**Remaining boundary.** The normalized q79 bridge derivative and all global worldsheet and string consistency tests must still be executed.

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## 1 Scope

This paper proves a local equivalence of quadratic diagnostics under explicit hypotheses. It does not prove:

- that a worldsheet is the unique extension of proto-spinor data;
- that MTT derives string theory, critical dimension, modular invariance, or a target-space compactification;
- that a local quadratic bridge extends to the full nonlinear theory; or
- that a matching number of carrier components identifies two global bundles.

## 2 The two configuration spaces

Let  $\mathcal{C}_{\text{ps}}$  be a Sobolev manifold of local proto-spinor, connection, and strain data near an aligned background  $u_*$ . A tangent vector is written

$$u = (u_{\text{circ}}, u_{\text{lens}}, u_{\text{nil}}) \in E_1 \oplus E_2 \oplus E_3,$$

where the labels denote operator/carrier roles. In the selected q79 interface their target ranks are 1, 2, 3.

Let  $\mathcal{C}_{\text{ws}}$  be a Sobolev manifold of worldsheet couplings near a background  $c_*$ ,

$$c = (G, B, \Phi, A, \mathcal{V}, \dots),$$

with worldsheet metric coupling  $G$ , antisymmetric field  $B$ , dilaton  $\Phi$ , gauge connection  $A$ , and any declared boundary or potential couplings  $\mathcal{V}$ . The ellipsis is not permission to omit fields needed by the chosen string model; all active couplings must be listed in an application.

**Definition 2.1** (Typed bridge map). A proto-spinor/worldsheet bridge on neighborhoods  $U \subset \mathcal{C}_{\text{ps}}$  and  $V \subset \mathcal{C}_{\text{ws}}$  is a  $C^2$  map

$$\mathcal{B} : U \longrightarrow V, \quad \mathcal{B}(u_*) = c_*.$$

Its derivative at the background is denoted  $L = D\mathcal{B}(u_*)$ .

This definition replaces an analogy between symbols by a map with a domain, codomain, and regularity class.

### 3 Local remainder control

Assume  $D\mathcal{B}$  is locally Lipschitz with constant  $M_B$ . Taylor's theorem gives

$$\mathcal{B}(u_* + u) = c_* + Lu + r_B(u), \quad \|r_B(u)\|_{\text{ws}} \leq \frac{M_B}{2} \|u\|_{\text{ps}}^2.$$

Let the two diagnostics have expansions

$$\begin{aligned} \mathcal{J}_{\text{ps}}(u_* + u) &= \mathcal{J}_{\text{ps}}(u_*) + \frac{1}{2} \langle u, H_{\text{ps}} u \rangle + R_{\text{ps}}(u), \\ \mathcal{J}_{\text{ws}}(c_* + v) &= \mathcal{J}_{\text{ws}}(c_*) + \frac{1}{2} \langle v, H_{\text{ws}} v \rangle + R_{\text{ws}}(v), \end{aligned}$$

with cubic bounds on the declared balls.

**Theorem 3.1** (Conditional quadratic bridge). *Let  $E_{\text{sel}} \subset T_{u_*} \mathcal{C}_{\text{ps}}$  be a closed selected subspace. Suppose  $L$  is injective on  $E_{\text{sel}}$  and*

$$L^* H_{\text{ws}} L = H_{\text{ps}} \quad \text{on } E_{\text{sel}}.$$

*If the displayed Taylor and cubic bounds hold, then there is  $C > 0$  such that for sufficiently small  $u \in E_{\text{sel}}$ ,*

$$|[\mathcal{J}_{\text{ws}}(\mathcal{B}(u_* + u)) - \mathcal{J}_{\text{ws}}(c_*)] - [\mathcal{J}_{\text{ps}}(u_* + u) - \mathcal{J}_{\text{ps}}(u_*)]| \leq C \|u\|_{\text{ps}}^3.$$

*Proof.* Insert  $v = Lu + r_B(u)$  into the worldsheet expansion. The pure quadratic term equals the proto-spinor quadratic term by the intertwining identity. Every term containing  $r_B$  is at least cubic because  $r_B = O(\|u\|^2)$ , and the two declared Taylor remainders are cubic. Their constants combine into  $C$ .  $\square$

**Corollary 3.2** (Local admissibility equivalence). *If both Hessians are coercive on the selected subspaces, then for sufficiently small perturbations the bridge preserves strict local admissibility and the Morse index on  $E_{\text{sel}}$ .*

This is the rigorous content of the earlier “shadow bridge.” It is local, quadratic to leading order, and conditional on a constructed  $\mathcal{B}$ .

### 4 Blockwise carrier map

Suppose the selected proto-spinor tangent carrier decomposes as

$$E_{\text{sel}} = E_1 \oplus E_2 \oplus E_3, \quad \dim(E_1, E_2, E_3) = (1, 2, 3),$$

and choose corresponding worldsheet coupling blocks  $W_1, W_2, W_3$ . A block-preserving derivative has form

$$L = L_1 \oplus L_2 \oplus L_3, \quad L_j : E_j \rightarrow W_j.$$

If

$$H_{\text{ps}} = \bigoplus_j H_{\text{ps},j}, \quad H_{\text{ws}} = \bigoplus_j H_{\text{ws},j}, \quad L_j^* H_{\text{ws},j} L_j = H_{\text{ps},j},$$

then the bridge theorem holds blockwise.

The roles may be read as follows, provided an application constructs the actual maps:

**Circle block.** Common  $U(1)$  phase/holonomy may map to a compact worldsheet scalar, Wilson line, or phase coupling.

**Lens block.** Finite/projective transport may map to orbifold or twisted sector data.

**Nil block.** Triangular anchoring or boundary termination may map to boundary couplings, filtration data, or a nilpotent differential.

These are typed possibilities, not a proof that the target worldsheet contains a literal circle, lens space, or Nil manifold.

## 5 The selected q79 target

The q79 degree-three cover supplies

$$\mathcal{H}_{\text{q79}} = L_{\text{shared}} \otimes (\mathcal{O} \oplus \mathcal{A}_0 \oplus \mathcal{A}), \quad \text{rank} = 1 + 2 + 3.$$

This gives a concrete source for the block ranks. It does not yet provide the bridge derivative  $L$ . To do so one must construct normalized q79 zero modes and evaluate how their metric,  $B$ -field, connection, and scalar variations enter the worldsheet sigma-model couplings.

The required same-source diagram is

$$\begin{array}{ccc} E_{\text{strain}} & \xrightarrow{\mathfrak{J}_{\text{WW} \rightarrow \text{q79}}} & \mathcal{H}_{\text{q79}} \\ \downarrow & & \downarrow D\mathcal{B}_{\text{q79}} \\ TC_{\text{ps}} & \xrightarrow{L} & TC_{\text{ws}}. \end{array}$$

Both routes must agree, and the covariant derivatives and Hessians must intertwine. Equality of ranks does not make the square commute.

## 6 Why two-dimensionality is not forced

A two-dimensional worldsheet is a distinguished encoding because local sigma-model couplings, conformal methods, and string propagation live there. But no theorem in the proto-spinor carrier alone excludes a worldline, a higher-dimensional defect, a lattice transfer system, or an operator-algebraic encoding. Uniqueness would require a classification theorem with assumptions strong enough to single out a two-dimensional local conformal theory.

Accordingly, this paper treats the worldsheet as a selected target-compatible extended chart, not as an inevitable consequence of failed three-dimensional closure.

## 7 Spinorial and twistor chart languages

The internal proto-spinor is conditional on a lift of rank-three orientation data. A spacetime Weyl or Dirac spinor additionally requires a Lorentzian spin bundle and Dirac operator. A Majorana chart requires a compatible antilinear real structure and charge constraints. A twistor chart requires the relevant conformal and integrability conditions.

These descriptions may coexist on one coherent slab and be related by typed maps. Their coexistence demonstrates representational flexibility, not that one chart dynamically produces the others.

## 8 Worldsheet consistency gates

Even an exact local quadratic bridge does not define a consistent string theory. A selected application must address:

1. vanishing beta functions or a controlled off-shell worldsheet RG flow;
2. Weyl anomaly and criticality, including ghosts and central charge;
3. modular invariance and the spin-structure/GSO sum where applicable;
4. gauge and gravitational anomaly cancellation;
5. boundary conditions, open/closed consistency, and tadpoles;
6. unitarity and the physical state conditions; and
7. the map from worldsheet beta functions to target-space field equations, including the order in  $\alpha'$  and truncation error.

For heterotic Fu–Yau backgrounds, satisfaction of the Hull–Strominger system is powerful target-space evidence but does not automatically prove all-order worldsheet conformal invariance.

## 9 Relation to finite SM calculations

The selected finite-algebra and renormalized-SM packets establish equivalence at the declared one-shared-physical-primitive/profile standard. Those packets do not require a worldsheet derivation and therefore remain valid independently of this bridge. Conversely, a worldsheet bridge cannot promote measured profile input into strict no-knob source selection unless its normalized overlap integrals and transport scheme emit the values independently.

## 10 Execution protocol

To promote the bridge beyond its current conditional status:

1. choose one selected q79 Fu–Yau background and its HYM connection;
2. compute normalized harmonic/Dirac representatives for the three carrier blocks;
3. differentiate the sigma-model coupling map to obtain  $L$ ;
4. compute  $H_{\text{ps}}$  and  $H_{\text{ws}}$  from the same action and verify  $L^*H_{\text{ws}}L = H_{\text{ps}}$  with interval or exact certificates;
5. bound  $D^2\mathcal{B}$  and both cubic remainders on an explicit ball; and
6. run the independent worldsheet consistency gates.

## 11 Conclusion

The proto-spinor/worldsheet relation is now a precise, testable local theorem. The bridge preserves quadratic admissibility when its derivative intertwines the Hessians, with a cubic error bound. The selected q79 carrier gives the right finite block target, while the actual normalized derivative and global worldsheet consistency remain to be computed. This is meaningful progress without claiming that the existence of a proto-spinor forces string theory.

## References

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